CSE 6013
Computational Linguisti
Lecture-I

Computational Linguistics:

It builds on the theory and practice of multiple fields (linguistics, computer science and statistics) to design computer applications that involve the automatic processing of natural language speech or text by machines.

Course Outlines:

1. Finite state automata
2. Regular expression
3. Context free grammar
4. Parsing
5. Morphology
6. phonetics and Speech Recognition
7. N-gram
8. TTS (Text to Speech)
9. Semantics and Syntax
10. Machine Translation

Automata theory:

Alphabet: An alphabet is a finite, nonempty set of symbols.
Σ is alphabet symbol.

Example:

binary alphabet, $\Sigma = \{0, 1\}$
set of all lower-case letters, $\Sigma = \{a, b, \ldots, z\}$

String:

A string is a finite sequence of symbols chosen from alphabet.

Example: 01101 is a string, where $\Sigma = \{0, 1\}$
computer is a string, where $\Sigma = \{a, b \ldots z\}$

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Empty string:

The empty string is the string with zero occurrences

ϵ is empty string.

Length of a string:

The length of a string is the number of positions for symbols in the string.

string 01101, length of 01101 = $|01101| = 5$

string computer, length of computer = $|\text{computer}| = 8$

Empty string ϵ, length of $\epsilon = |\epsilon| = 0$

Powers of an Alphabet

If Σ is an alphabet, we can express the set of all strings of a certain length from that alphabet by using an exponential notation.

Σ^k = the set of strings of length k.

Example:

$\Sigma^0 = \{\epsilon\}$

$\Sigma^1 = \{0, 1\}$

$\Sigma^2 = \{00, 01, 10, 11\}$

$\Sigma^3 = \{000, 001, 010, 011, 100, 101, 110, 111\}$

Σ^* = The set of all strings

$= \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \ldots$

$= \{\epsilon\} \cup \{0,1\} \cup \{00, 01, 10, 11\} \cup \ldots$

$= \{\epsilon, 0, 1, 00, 01, 10, 11, \ldots\}$

Σ^+ = The set of all non-empty strings

$= \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \ldots$

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